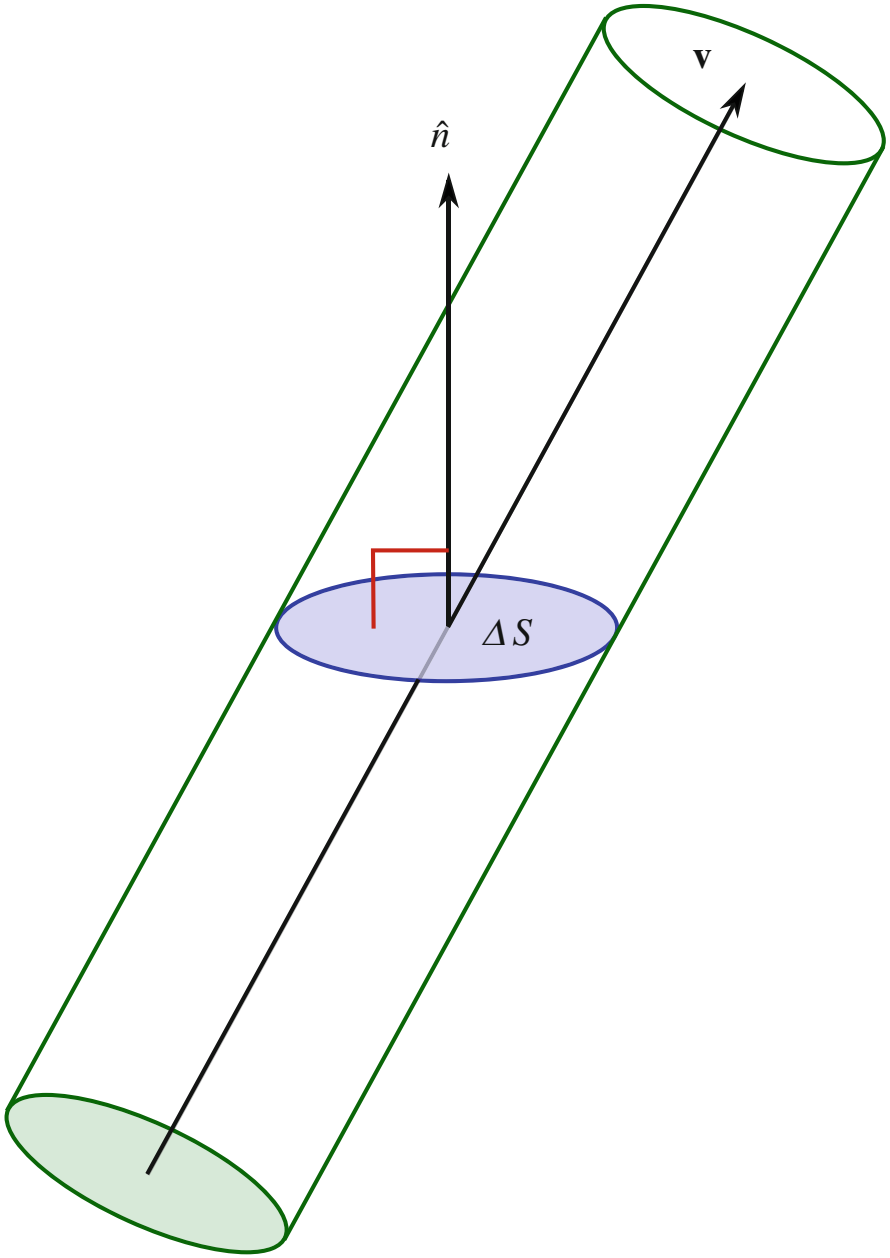
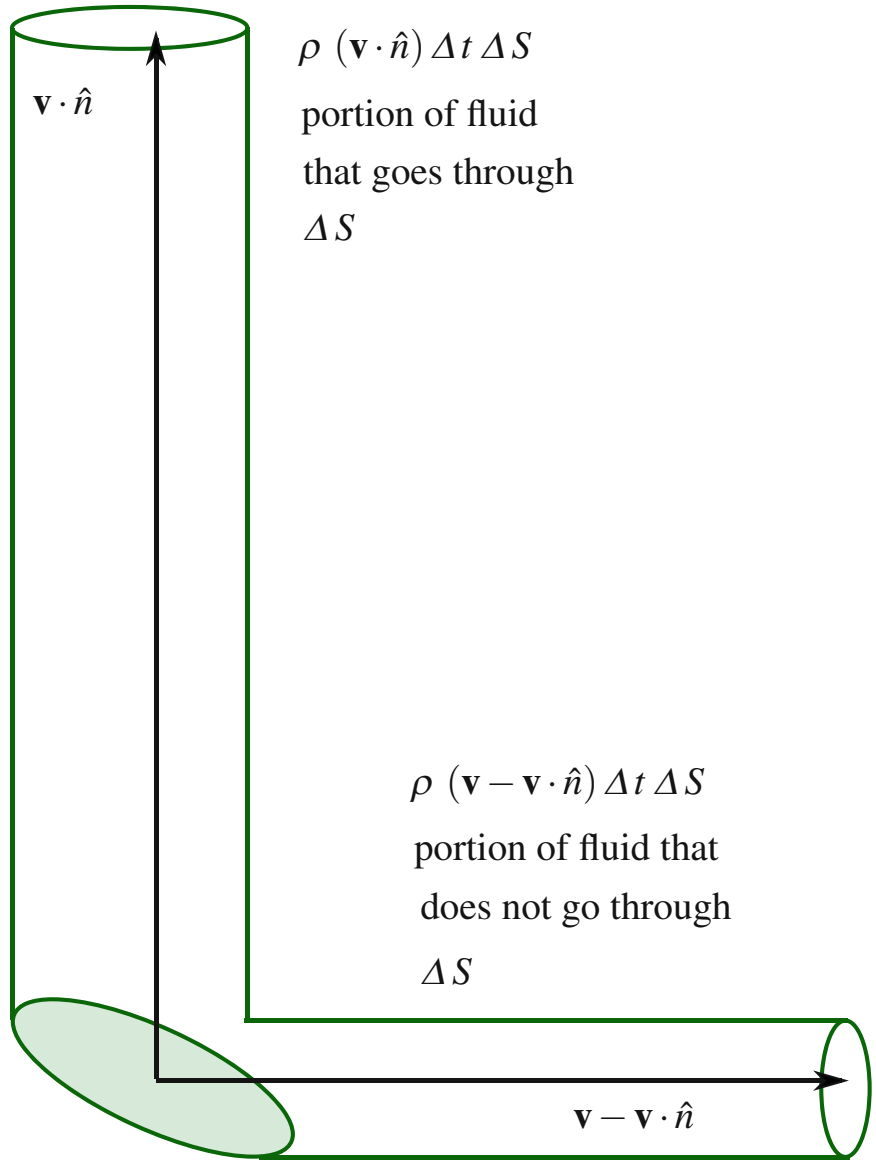

$\mathbf{v}$
$\hat{n}$
ΔS


$\mathbf{v} \cdot \hat{n}$
$\rho\ (\mathbf{v} \cdot \hat{n})\, \Delta t\, \Delta S$
portion of fluid
that goes through
ΔS
$\rho\ (\mathbf{v} - \mathbf{v} \cdot \hat{n})\, \Delta t\, \Delta S$
portion of fluid that
does not go through
ΔS
$\mathbf{v} - \mathbf{v} \cdot \hat{n}$